Smart Waste Management Bin
-WasteWise-

Project Proposal

Team Name	WasteWise
Category	Smart Environment & Waste Management
Theme	Human-Centric AI for Sustainable Waste Management

Problem Definition

Introduction

Rapid urbanization and population growth have increased solid waste generation in public places such as universities, parks, beach sides and transportation hubs. Traditional bins rely on manual monitoring, often leading to overflowing waste, odors, and unsanitary conditions, which harm public health and urban hygiene. Effective waste segregation is also challenging due to lack of awareness, disabilities, time constraints, and user confusion, causing recyclables and organic waste to mix and reducing recycling efficiency.

Industry 5.0 promotes human-centric, sustainable solutions. Integrating IoT and AI into waste management can make systems smarter, more inclusive, and efficient, while supporting the United Nation Sustainable Development Goals.

Problem Analysis

The current waste management system faces several critical challenges:

- Waste bins are not monitored in real time, leading to delayed collection, overflowing garbage, foul odors and unhygienic environments.
- Improper waste segregation is common, as users often struggle to identify the correct disposal due to confusion, lack of awareness, or physical disabilities such as visual impairment.
- Manual inspection and collection of waste bins require significant amounts of time, labor, and operational costs, making the process inefficient and unreliable.
- Many existing smart waste management solutions are complex and expensive, making them unsuitable for small-scale locations, educational institutions, or low-budget deployments.
- The absence of user feedback mechanisms results in careless waste disposal, reducing recycling efficiency and increasing environmental impact.

Proposed Solution

Proposed Product

The proposed solution, **WasteWise,** is an IoT-based smart waste management bin designed as a simple, eco-friendly, and inclusive prototype that enhances waste handling through automation and AI-driven decision-making.

WasteWise integrates multiple sensors and embedded systems, including Ultrasonic Sensors, Gas Sensors, Weight Sensors, MG995 Motor, Micro Servo Motors, OV2640 Camera Module, LCD Display, Arduino Uno & ESP32 microcontroller, LEDs, a Buzzer, Bluetooth connectivity, and 5V Mini Solar Panel.

Together these elements empower **AI-powered waste sorting, intelligent fill-level prediction, hazardous gas & odor monitoring, and a human-centric user interaction system,** creating a sustainable waste management experience.

Uniqueness of the Solution

WasteWise differentiates itself from conventional and existing smart waste management systems through the following innovative features:

- **AI-driven waste classification combined with an adaptive rotating bin system**, allowing automatic routing of waste into the correct compartment without requiring user effort.
- A **human-centric design approach** that reduces cognitive and physical effort, making the system accessible to children, elderly individuals, and people with disabilities.
- **Edge AI using TinyML**, enabling real-time processing on the device without relying on cloud infrastructure, improving privacy, speed, and reliability.
- **Sustainability energy usage** through integrated solar power and battery backup, aligning the system with environmental sustainability goals.
- A **mobile application interface** that provides real-time updates on bin fill levels and system status, improving waste collection efficiency and reducing operating costs.
- Integrating **odor monitoring** to enhance hygiene and user comfort in public spaces.

Technical Overview and Implementation

Technical Details

WasteWise is designed as an intelligent and human-centric smart waste management system that automates waste segregation while minimizing user effort. The system focuses on improving recycling efficiency by accurately classifying and routing waste into appropriate compartments.

- The smart bin consists of **three internal compartments** dedicated to **organic, plastic and paper waste**. At the top of the bin, an automatic lid mechanism is installed to allow hands-free disposal. When a user approaches the bin, an **ultrasonic sensor** detects the presence of an object and triggers the lid to open using a micro servo motor.
- Once waste is placed into the bin, an **AI-powered camera module (OV2640)** captures an image of the item. This image is processed locally on the **ESP32 microcontroller** using a **TinyML-based image classification model** to identify the waste category. Based on the classification result, an **adaptive rotating mechanism** controlled by an MG995 motor aligns the correct internal compartment beneath the disposal opening.
- A **small LCD display** provides real-time visual feedback, such as system status messages and disposal confirmations. Additionally, the system continuously monitors bin conditions using **weight sensors** to measure fill levels and **gas sensors** to detect odor or harmful gas buildup caused by waste decomposition.
- Through **Bluetooth connectivity**, WasteWise sends real-time alerts to connected mobile applications. Notifications are triggered when bin capacity reaches predefined thresholds or when gas levels exceed safe limits, helping prevent health risks and maintaining hygiene in public spaces.
- To ensure sustainability, the entire system is powered using **solar energy**, supported by a battery backup and charge controller. This allows WasteWise to operate efficiently with minimal dependence on external power sources, reducing energy consumption and supporting environmentally responsible operation.

User Scenario

Kasun is a visually impaired individual who finds it difficult to identify and use conventional waste segregation bins. In public spaces, the lack of accessibility and clear guidance often forces him to depend on others for assistance.

When Kasun approaches the WasteWise smart bin, the ultrasonic sensor detects his presence and automatically opens the lid. He places the waste item into the bin without needing to identify the waste category. The AI-powered system classifies waste and routes it to the correct internal compartment using the adaptive rotating mechanism.

Audio and visual feedback from the system assures Kasun that the disposal process has been completed successfully. He does not need to interact with multiple bins or rely on external help. Additionally, the mobile application sends notifications to responsible users or authorities when the bin becomes full or when odor levels increase.

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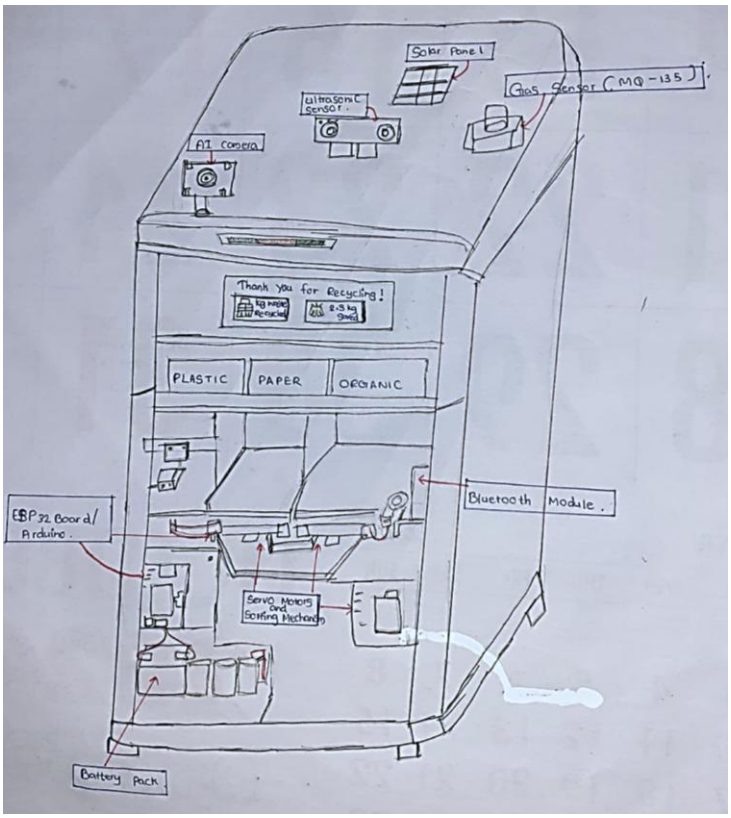
Additional Information



This is a simple structure we drew to get a better idea about the final product going to look like.